PULLBACK COMMUTES WITH DERIVED LIMITS OF SURJECTIVE SYSTEMS OF MODULES OVER A PROPER FLAT FINITELY PRESENTED SCHEME UP TO QUASI-COHERENT COHERATOR

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Allowed standard prerequisites (editorial).

- D_QCoh of a quasi-compact quasi-separated scheme has a perfect generator. invoked at proof lines 7805-7807. Reduces the assertion to perfect test objects E.
- Derived limits of quasi-coherent modules agree with R lim, and the functor R Hom_X(E, -) into D(R) commutes with products hence with derived limits. invoked at proof lines 7808-7818. Moves the limit out of the Hom.
- The dual perfect complex computes Hom, the base change lemma computes R Gamma of the dual tensor a pulled back module as a tensor product over R, and R Gamma(X, E^vee) is perfect for proper f with E perfect. invoked at proof lines 7820-7831. Translates each term into a tensor product of R-modules.
- A pseudo-coherent complex of modules tensored into a surjective inverse system satisfies the limit isomorphism, by the More on Algebra pseudo-coherent tensor limit lemma. invoked at proof lines 7833-7838. Concludes the derived-limit exchange.

Source reference. Original theorem 09IS: sources/stacks-perfect/pinned-original-source.tex, source lines 4111–4111. Cited at proof lines 7805-7806 to reduce the test objects to perfect E; the Hom reformulation is inside the counted span.

Source reference. Original lemma 0A0J: sources/stacks-perfect/pinned-original-source.tex, source lines 319–319. Cited at proof lines 7808-7809 to replace the limit by R lim; the derived-limit exchange is inside the counted span.

Source reference. Original section 0B6A: sources/stacks-chow/tags.txt, source lines 14063–14063. Cited at proof lines 7811-7812 as the reference for the functor R Hom_X(E, -); its product commutation use is inside the counted span.

Date: stacks-project:perfect.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 08DQ: sources/stacks-chow/tags.txt, source lines 10645–10645. Cited at proof lines 7819-7820 for the dual E^vee; the tensor rewriting is inside the counted span.

Source reference. Original lemma 08EU: sources/stacks-perfect/pinned-original-source.tex, source lines 5269–5269. Cited at proof lines 7830-7831 to compute R Hom as a tensor product over R; the limit exchange is inside the counted span.

Source reference. Original lemma 0B91: sources/stacks-perfect/pinned-original-source.tex, source lines 7742–7742. Cited at proof lines 7832-7833 for the perfectness of R Gamma(X, E^vee); the pseudo-coherent limit argument is inside the counted span.

Source reference. Original lemma 0D2L: sources/stacks-chow/tags.txt, source lines 16386–16386. Cited at proof lines 7835-7838 to conclude the derived-limit exchange; the assembly is inside the counted span.

Lemma 0.2. *Let R be a ring. Let X be a scheme and let $f : X \to \mathrm{Spec}(R)$ be proper, flat, and of finite presentation. Let (M_n) be an inverse system of R-modules with surjective transition maps. Then the canonical map*

$$\mathcal{O}_X \otimes_R (\lim M_n) \longrightarrow \lim \mathcal{O}_X \otimes_R M_n$$

induces an isomorphism from the source to DQ_X applied to the target.

Proof. The statement means that for any object E of $D_{QCoh}(\mathcal{O}_X)$ the induced map

$$\mathrm{Hom}(E, \mathcal{O}_X \otimes_R (\lim M_n)) \longrightarrow \mathrm{Hom}(E, \lim \mathcal{O}_X \otimes_R M_n)$$

is an isomorphism. Since $D_{QCoh}(\mathcal{O}_X)$ has a perfect generator (Theorem 09IS) it suffices to check this for perfect E. By Lemma 0A0J we have $\lim \mathcal{O}_X \otimes_R M_n = R\lim \mathcal{O}_X \otimes_R M_n$. The exact functor $R\mathrm{Hom}_X(E, -) : D_{QCoh}(\mathcal{O}_X) \to D(R)$ of Cohomology, Section 0B6A commutes with products and hence with derived limits, whence

$$R\mathrm{Hom}_X(E, \lim \mathcal{O}_X \otimes_R M_n) = R\lim R\mathrm{Hom}_X(E, \mathcal{O}_X \otimes_R M_n)$$

Let $E^\vee$ be the dual perfect complex, see Cohomology, Lemma 08DQ. We have

$$R\mathrm{Hom}_X(E, \mathcal{O}_X \otimes_R M_n) = R\Gamma(X, E^\vee \otimes^{\mathbf{L}}_{\mathcal{O}_X} Lf^* M_n) = R\Gamma(X, E^\vee) \otimes^{\mathbf{L}}_R M_n$$

by Lemma 08EU. From Lemma 0B91 we see $R\Gamma(X, E^\vee)$ is a perfect complex of R-modules. In particular it is a pseudo-coherent complex and by More on Algebra, Lemma 0D2L we obtain

$$R\lim R\Gamma(X, E^\vee) \otimes^{\mathbf{L}}_R M_n = R\Gamma(X, E^\vee) \otimes^{\mathbf{L}}_R \lim M_n$$

as desired. □